

The background features a light gray field with faint, overlapping circular patterns. Some of these circles have small tick marks and numbers along their perimeters, resembling a technical or scientific diagram. The numbers visible include 150, 160, 170, 180, 190, 210, 220, 230, 240, 250, and 260.

Genetic Algorithm

Presentation Outline:

- 1) Introducing the 4-Queen problem
- 2) **Activity**: Solving 4-Queen problem using artifacts
- 3) Solution of 4-Queen problem in Backtracking approach
- 4) Demerits of Backtracking approach
- 5) Introducing 8-Queen problem
- 6) Discussion on Genetic Algorithm
- 7) Solution of 8-Queen problem using GA
- 8) Conclusion



The 4-Queen Problem

Once upon a time, there was a great king in India. However, it was a matter of shame that he had 4 Queens. The Queens were so arrogant and they didn't even want to see one another. Therefore, the King built a castle of 4×4 rooms. However, he couldn't find a way to place the 4 Queens in 4 separate rooms, so that they couldn't see each other.

Would, you please help the King to place the Queens? Avoid placing two Queens in a same row, same column and even same diagonal rooms.



Solution of the 4-Queen Problem Using Backtracking Approach

Therefore , the king called Professor John Holland of the University of Michigan to solve the 4-Queen problem. And Professor solved the 4-Queen problem in backtracking approach.





The 5-Queen Problem



One month later, Professor received a call from the great King to solve his 5-Queen problem. Professor, solved the 5-Queen problem in backtracking approach.



Solution of the 5-Queen Problem Using Backtracking Approach





6-Queen Problem

John Holland introduced **Genetic Algorithm (GA)**

Darwin's theory of evolution



Fortunately, one month later, the King requested the professor to solve 6-Queen problem. The professor thought that the King may request him to solve 16-Queen problem within next 10 months.

Backtracking approach will not be efficient to solve the 8 or 16-Queen problems.

Therefore, professor invented Genetic Algorithm to solve the n-Queen problem.

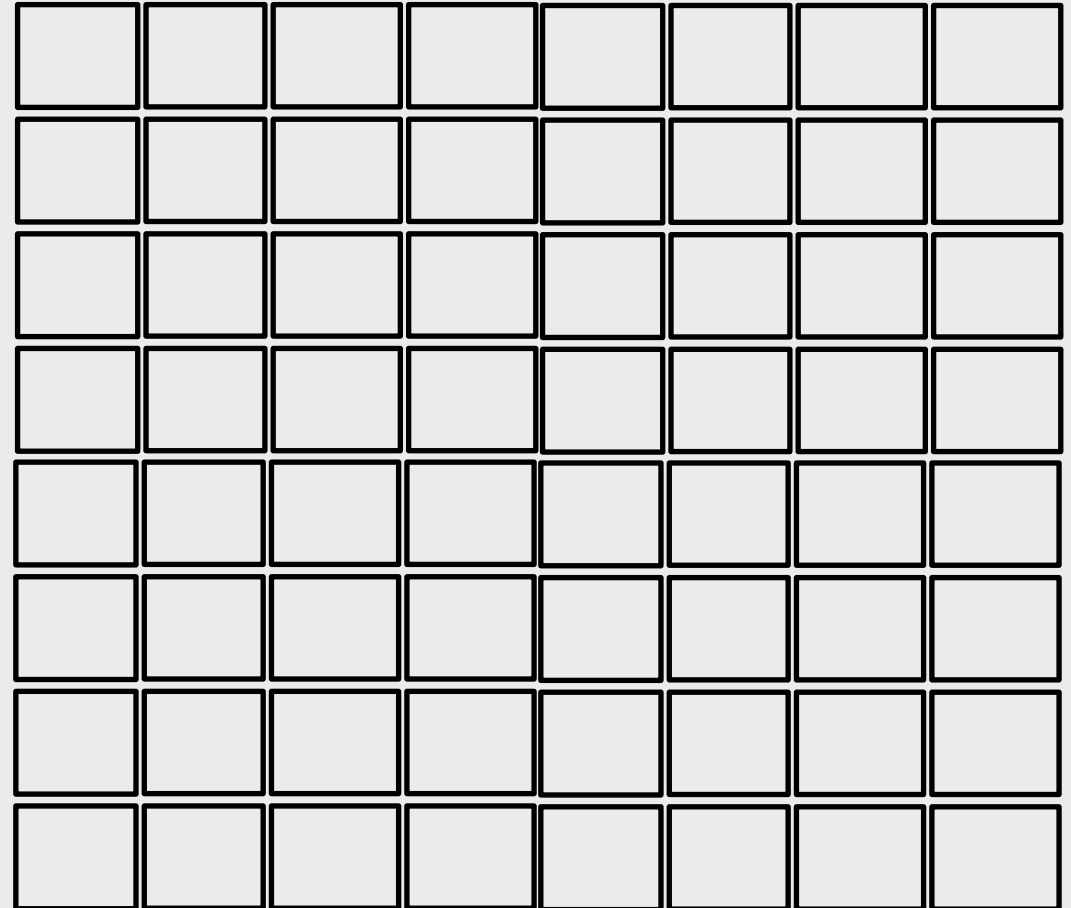


8-Queen Problem



John Holland introduced **Genetic Algorithm (GA)**

Darwin's theory of evolution



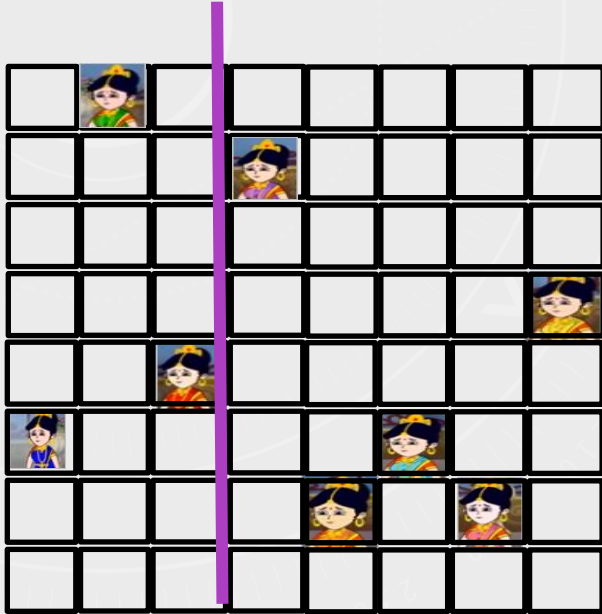
Introduced in the 1970s by John Holland at University of Michigan

- ▶ begin with k randomly generated states (population)
- ▶ each state (individual) is a string over some alphabet (chromosome)
- ▶ fitness function (bigger number is better)
- ▶ crossover
- ▶ mutate (evolve?)



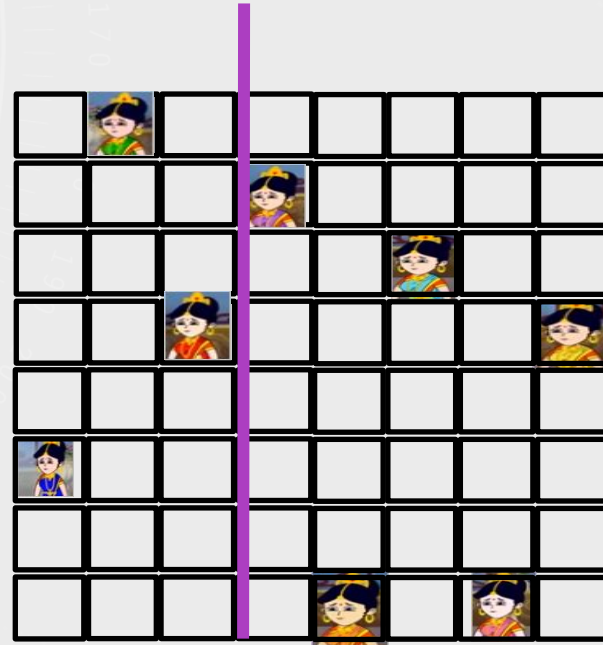
Formulation of Genetic Algorithm

John Holland introduced **Genetic Algorithm (GA)**
Darwin's theory of evolution



3 8 4 7 2 3 2 5

Fitness=28-7=21



3 8 5 7 1 6 1 5

Fitness=28-4=24

Fitness function: number of non-attacking pairs of queens

Maximum number of pairs: $8 \times 7/2 = 28$

[Q1 Q2]

[Q1 Q3]

[Q1 Q4]

[Q1 Q5]

[Q1 Q6]

[Q1 Q7]

[Q1 Q8]

.....

[Q8 Q7]

Chromosome of Father: 3 8 4 7 2 3 2 5

Chromosome of Mother: 3 8 5 7 1 6 1 5

Pseudo-code of GA:

START

 Generate the initial population

 Compute fitness

 REPEAT

 Selection

 Crossover

 Mutation

 Compute fitness

 UNTIL population has converged

STOP

Crossover:

Chromosome of Father:

3 8 4 7 2 3 2 5

Chromosome of Mother:

3 8 5 7 1 6 1 5

Crossover point

3 8 4 7 2 3 2 5

3 8 5 7 1 6 1 5

2 4 4 1 5 1 2 4

3 2 5 4 3 2 1 3

Chromosome of Father:

3 8 4 7 2 3 2 5

Chromosome of Mother:

3 8 5 7 1 6 1 5

Offspring 1:

3 8 4 7 1 6 1 5

Offspring2:

3 8 5 7 2 3 2 5

Mutation:

Before Mutation:

Offspring 1: 3 8 4 7 1 6 1 5

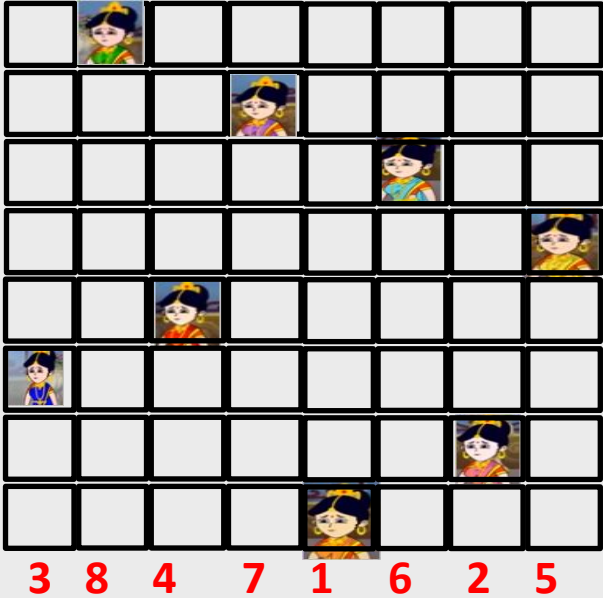
Offspring2: 3 8 5 7 2 3 2 5

After Mutation:

Offspring 1: 3 8 4 7 1 6 2 5

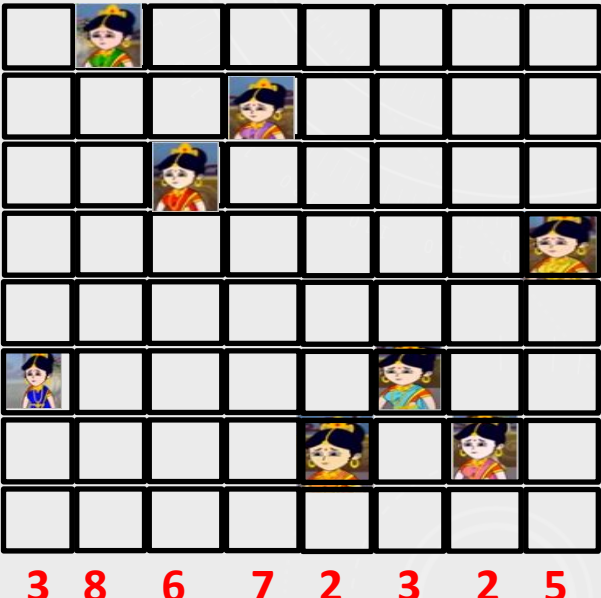
Offspring2: 3 8 6 7 2 3 2 5

Offspring 1:



Fitness=28-0=28

Offspring2:



Fitness=28-5=23

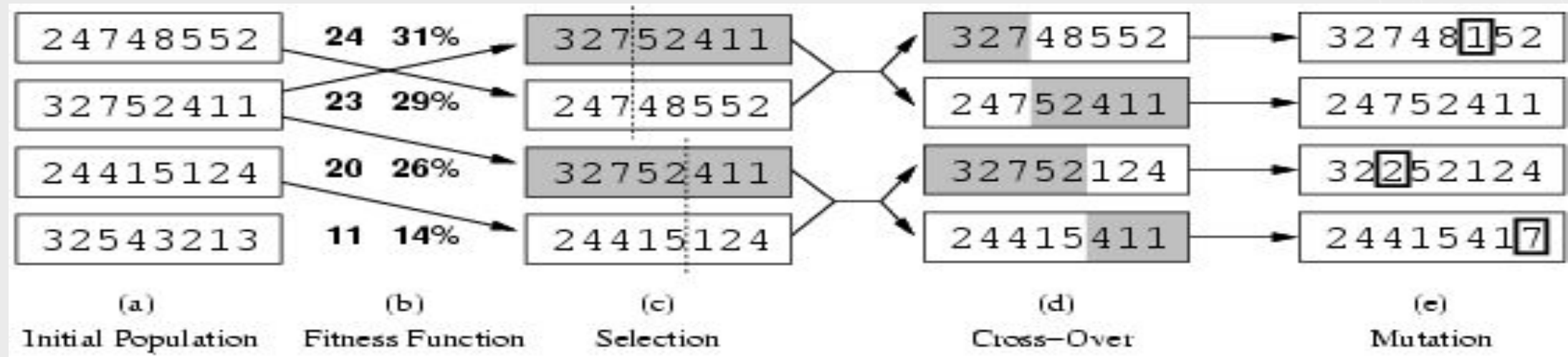
Represent states and compute fitness function.

24748552	24
32752411	23
24415124	20
32543213	11
	<u>77</u>

(a)

Initial Population

GENETIC ALGORITHMS

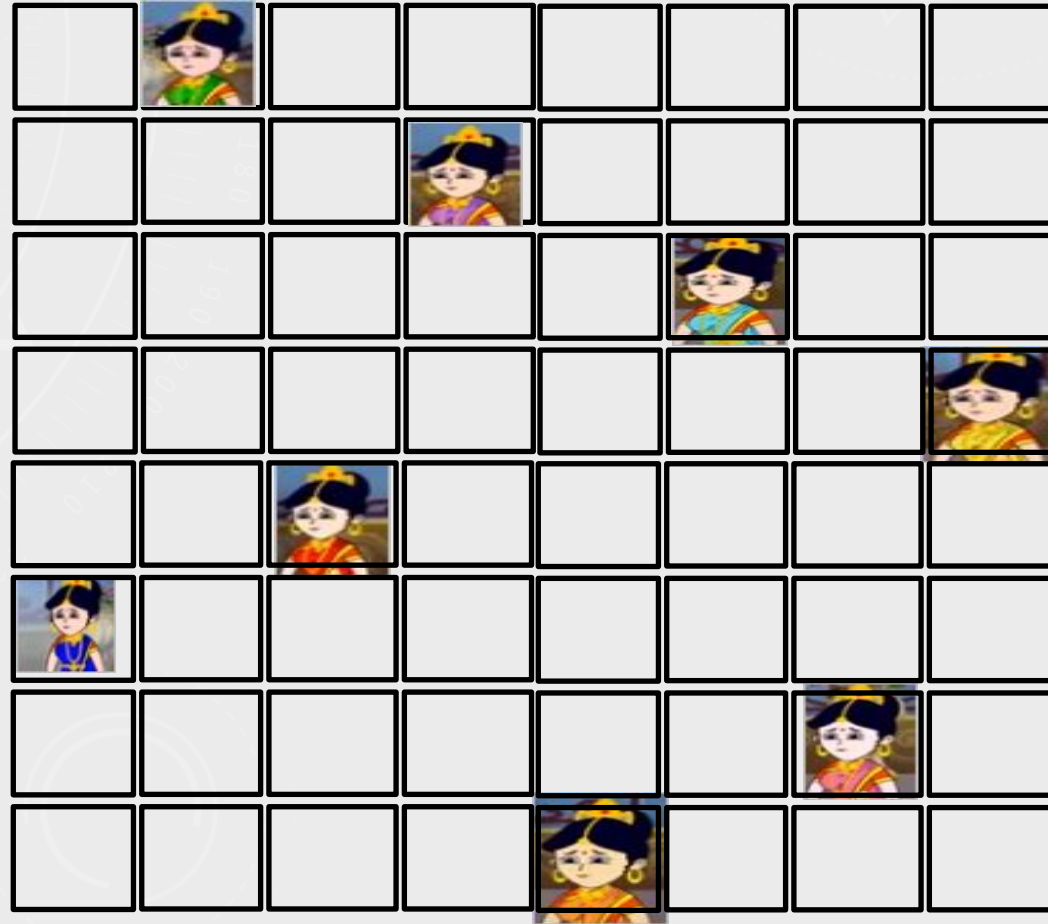


- Fitness function: number of non-attacking pairs of queens (min = 0, max = $8 \times 7/2 = 28$)
 $24/(24+23+20+11) = 31\%$
 $23/(24+23+20+11) = 29\%$ etc



Solution of 8-Queen Problem using Genetic Algorithm

John Holland introduced **Genetic Algorithm (GA)**
Darwin's theory of evolution

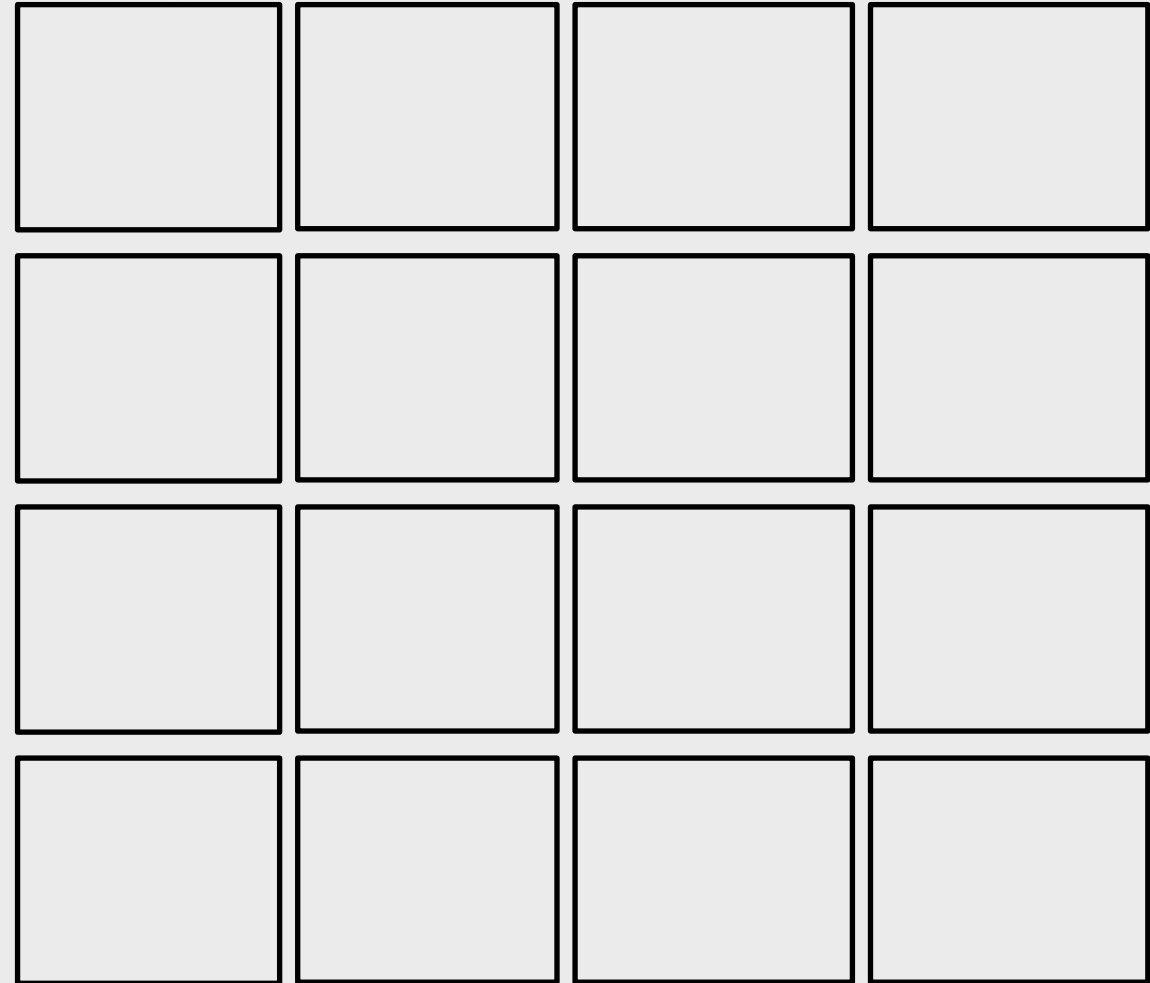




The 4-Queen Problem

Fitness function: number of non-attacking pairs of queens

What is the Maximum fitness value: ????





4-Queen Problem Using Backtracking Approach

Therefore , the king called Professor John Holland of the University of Michigan to solve the 4-Queen problem. And solved the 4-Queen problem in backtracking approach.





Solution of the 4-Queen Problem Using GA



Initial Population

Conclusion

Application areas of GA:

- ❑ Game programming
- ❑ Cloud resource allocation
- ❑ Job scheduling of operating systems
- ❑ Channel assignment in communication system
- ❑ Combinatorial optimization
- ❑ Creative design (NASA antenna)
- ❑ Operational research

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RESEARCH

A Dynamic Scheduling Method for Collaborated Cloud with Thick Clients

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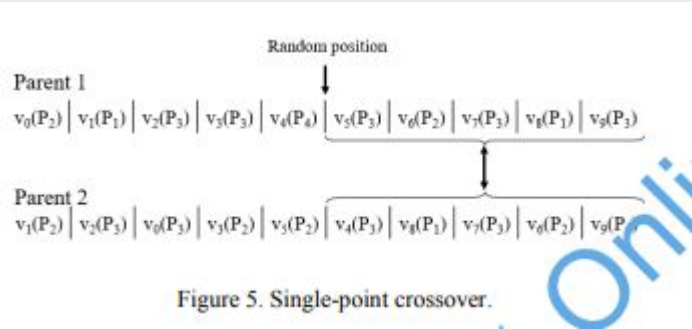


Figure 5. Single-point crossover.

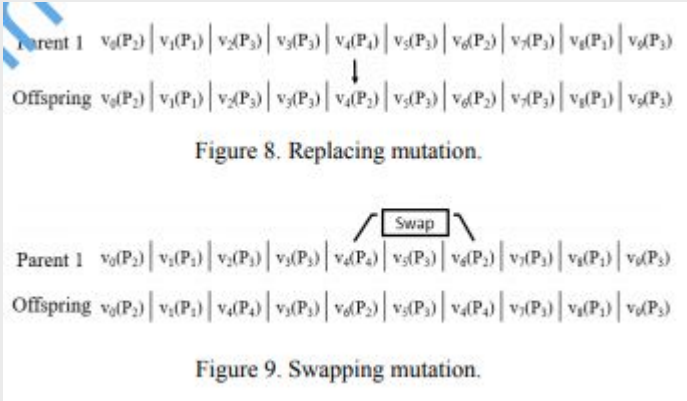


Figure 8. Replacing mutation.

Figure 9. Swapping mutation.